

HW 1.

Due: Sep. 30, 2026 at 3:30pm.

Please use python as your programming language.

Problem 1. Describe the column space of $A = \begin{bmatrix} v & w & v + 3w \end{bmatrix}$ where $v, w \in \mathbb{R}^3$. Describe the null space of A . What are the dimension of the column and null space of A ?

Problem 2. Let $A \in \mathbb{C}^{m \times m}$ be hermitian.

- a) Prove that all eigenvalues of A are real.
- b) Prove that if x and y are eigenvectors corresponding to distinct eigenvalues, then x and y are orthogonal.

Problem 3. If u and v are vectors in $\mathbb{R}^m$, the matrix $A = I + uv^T$ is known as a rank-one perturbation of the identity. Show that if A is nonsingular, then its inverse has the form $A^{-1} = I + \alpha uv^T$ for some scalar α , and give an expression for α . For what u and v is A singular? If it is singular, what is $\text{null}(A)$?

Problem 4. Show that if E is an outer product $E = uv^T$ for $u, v \in \mathbb{R}^m$, then $\|E\|_2 = \|u\|_2 \|v\|_2$. Is the same true for the Frobenius norm, i.e. $\|E\|_F = \|u\|_F \|v\|_F$? Prove it or give a counterexample.

Problem 5. Consider

$$A = \begin{pmatrix} 4 & -3 & 2 \\ -1 & 0 & 5 \\ 2 & 6 & -2 \end{pmatrix}.$$

Find $\|A\|_1$, $\|A\|_2$, $\|A\|_\infty$ and $\|A\|_F$.

Problem 6. Consider the matrix

$$A = \begin{pmatrix} -2 & 11 \\ -10 & 5 \end{pmatrix}.$$

- a) Determine, on paper, a real SVD of A in the form $A = U\Sigma V^T$. The SVD is not unique, so find the one that has the minimal number of minus signs in U and V .
- b) List the singular values, left singular vectors, and right singular vectors of A . Draw a careful, labeled picture of the unit ball in $\mathbb{R}^2$ and its image under A , together with the singular vectors, with the coordinates of their vertices marked.
- c) What are $\|A\|_1$, $\|A\|_2$, $\|A\|_\infty$ and $\|A\|_F$?
- d) Find A^{-1} not directly, but via the SVD.

- e) Find the eigenvalues λ_1, λ_2 of A .
- f) Verify that $\det A = \lambda_1 \lambda_2$ and $|\det A| = \sigma_1 \sigma_2$.
- g) What is the area of the ellipsoid onto which A maps the unit ball of $\mathbb{R}^2$?

Problem 7. Consider

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

- a) Show that the eigenvalues and corresponding eigenvectors of A are as follows:

$$\lambda_k = 4 \sin^2\left(\frac{k\pi}{10}\right), \quad v_{k,j} = \sin\left(\frac{kj\pi}{5}\right) \quad k = 1, 2, 3, 4, \quad j = 1, 2, 3, 4$$

where $v_k = [v_{k,1} \ v_{k,2} \ v_{k,3} \ v_{k,4}]^T$ is the eigenvector associated with λ_k . What are a SVD of A ?

- b) Write a program of SVD where you are allowed to use the “numpy.linalg.eig” function to compute the eigenvalue and eigenvectors. Do not call the “numpy.linalg.svd” function to compute SVD of A . The purpose of this problem is to get you familiar with the python programming in the context of scientific computing.
- c) Compare your results from b) with the svd from a).
- d) (Generalization) Consider a generalization of A to $\mathbb{R}^{n \times n}$:

$$A = 2 * \text{np.diag}(\text{np.ones}(n)) - \text{np.diag}(\text{np.ones}(n-1), -1) - \text{np.diag}(\text{np.ones}(n-1), 1)$$

What are the eigenvalues of $A \in \mathbb{R}^{n \times n}$?

Apply your program implemented in b) to A with $n = 12$ to find singular values of A .